Gas Laws

41. (d)

42. (b) Ideal gases cannot be liquefied easily.

43. (a) For *m gm* of gas $PV = mrT$

Substituting the given and known values

$$r = \frac{1.015\times10^5\times10^{-3}}{1.293\times273} = 0.29\ J/K\text{-}gm$$

44. (c) We have $\frac{PV}{T} = \frac{P_0 m}{d_0 T_o}$

$$\therefore m = \frac{PVd_0T_0}{TP_o} = \frac{760\times10^{-3}\times1.2\times10^3\times273}{76\times294} = 1.13gm$$

45. (a) The value of universal gas constant is approve $2\frac{cal}{mole\text{-}Kelvin}$

46. (a) $PV = NKT \Rightarrow N = \frac{PV}{KT}$

$$= \frac{(1.64\times10^{-3}\times1.01\times10^5)\times(1\times10^{-6})}{1.38\times10^{-23}\times200} = 6.02\times10^{16}$$

47. (d) $PV = NkT \Rightarrow \frac{N_A}{N_B} = \frac{P_A\ V_A}{P_B\ V_B}\times\frac{T_B}{T_A}$

$$\Rightarrow \frac{N_A}{N_B} = \frac{P\times V\times(2T)}{2P\times\frac{V}{4}\times T} = \frac{4}{1}$$

48. (d) $PV = \mu RT = CT$

$\Rightarrow C = \mu R$, depends upon mass of gas.

49. (c) $P_1V_1 = P_2V_2 \Rightarrow 4\times V = 10\times1 \Rightarrow V = 10/4 = 2.5cc$

50. (d) $PV = kT \Rightarrow P\left(\frac{m}{\rho}\right) = kT \Rightarrow \rho = \frac{Pm}{kT}$

51. (c) $PV = \mu RT$ (Gas equation) $\Rightarrow PV \propto T$.

52. (c) $\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2} \Rightarrow \frac{1 \times 500}{300} = \frac{0.5 \times V_2}{270} \Rightarrow V_2 = 900m^3$

53. (c) $PV = \mu RT \Rightarrow P \propto \mu T$ ($\because$ V and R = constant)

$\Rightarrow \frac{P_2}{P_1} = \frac{\mu_2}{\mu_1} \times \frac{T_2}{T_1} \Rightarrow \frac{P_{He}}{P} = \frac{1}{1} \times \frac{2T}{T} \Rightarrow P_{He} = 2P.$

54. (b) $PV = \mu RT \Rightarrow \frac{P_1V_1}{P_2V_2} = \frac{\mu_1}{\mu_2} \Rightarrow \frac{50 \times 100}{100 \times V_2} = \frac{1}{2} \Rightarrow V_2 = 100\ ml.$

55. (d) Vander Waal's equation followed by real gases.

56. (c) For ideal gas, on free expansion there is no change in temperature. Hence $C_a = C_b$

57. (c)

58. (a) $\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2} \Rightarrow \frac{76 \times 1}{300} = \frac{152 \times 2}{T_2}$

$\Rightarrow T_2 = 1200K = 927°C$

59. (d) $PV = mrT$ Since P, V, $r \rightarrow$ remains same

Hence $m \propto \frac{1}{T} \Rightarrow \frac{m_1}{m_2} = \frac{T_2}{T_1} \Rightarrow \frac{13}{m_2} = \frac{(273+52)}{(273+27)} = \frac{325}{300}$

$\Rightarrow m_2 = 12\ gm$

i.e., mass released = 13*gm* – 12*gm* = 1*gm.*

60. (a) $PV = mrT = m\left(\frac{R}{M}\right)T$

$\Rightarrow V = \left(\frac{m}{M}\right)\frac{RT}{P} = \left(\frac{2.2}{44}\right) \times \frac{8.31 \times (273+0)}{2 \times (1 \times 10^5)}$

$= 5.67 \times 10^{-4} m^3$ = 0.56 litre.